A Counterexample to the Group-Ring Determinant Conjecture

Rewritten for readability from the OpenAI manuscript of the same title

September 23, 2026

1 Main result and proof at a glance

Conventions

Throughout, G is a discrete group and $R[G]$ is the group ring of *finite* sums $c = \sum_g c_g g$ with coefficients in a ring R. We write $[1]c = c_1$ for the identity coefficient and $c^* = \sum_g \overline{c_g}\, g^{-1}$. For a matrix, $(A^*)_{ij} = (A_{ji})^*$.

The left regular representation of G on $\ell^2(G)$ is $(L_g\xi)(h) = \xi(g^{-1}h)$. A matrix $A \in \mathrm{M}_n(\mathbb{C}[G])$ acts on $\ell^2(G)^n$ by

$$(T_A\xi)_i = \sum_j \sum_g (A_{ij})_g\, L_g\xi_j.$$

Since $g \mapsto L_g$ is a unitary representation, $A \mapsto T_A$ is a unital $*$-homomorphism: $T_{AB} = T_AT_B$ and $T_{A^*} = T_A^*$. Two consequences are used repeatedly. If $P = P^* = P^2$ (an *orthogonal projection*), then T_P is an orthogonal projection on $\ell^2(G)^n$. If A has an inverse in $\mathrm{M}_n(\mathbb{Q}[G])$, then T_A has a bounded inverse.

Let $e_i \in \ell^2(G)^n$ have δ_1 (the indicator of the identity) in component i and 0 elsewhere. The *unnormalized* trace is $\tau_n(T) = \sum_{i=1}^n \langle Te_i, e_i\rangle$, so $\tau_n(I_n) = n$. Because $\langle L_g\delta_1, \delta_1\rangle$ is 1 for $g = 1$ and 0 otherwise,

$$\tau_n(T_A) = \sum_{i=1}^n [1]A_{ii} =: \tau_n(A).$$

For $A \in \mathrm{M}_n(\mathbb{Z}[G])$ the probability spectral measure of $T_A^*T_A$ is

$$\mu_A(E) = \frac{1}{n}\sum_{i=1}^n \langle \mathbf{1}_E(T_A^*T_A)e_i, e_i\rangle \qquad (E \subseteq [0,\infty) \text{ Borel}).$$

A nonsingular integer matrix has $|\det| \geq 1$. The square-matrix case of the Determinant Conjecture [7, Conjecture 13.2] asserts the analogue

$$\int_{(0,\infty)} \log t \, d\mu_A(t) \geq 0 \tag{1.1}$$

for every G and A. The atom at 0 is omitted, and the integral may be $-\infty$. The conjecture as usually stated also allows rectangular matrices, so a square counterexample refutes it. When T_A is invertible, the Fuglede–Kadison determinant [4] over the group von Neumann algebra $\mathcal{N}(G)$ is defined by

$$\log \det_{\mathcal{N}(G)}(T_A) = \tfrac{1}{2}\tau_n\big(\log(T_A^*T_A)\big) = \tfrac{n}{2}\int_{(0,\infty)} \log t \, d\mu_A(t). \tag{1.2}$$

The second equality holds because $\langle f(T_A^* T_A)e_i, e_i\rangle$ is the integral of f against the i-th summand of $n\mu_A$.

Comparison with [7]. That reference uses right multiplication on row vectors. Conjugating by the unitary $\delta_g \mapsto \delta_{g^{-1}}$ in each component turns L_g into $\delta_h \mapsto \delta_{hg^{-1}}$. Hence it turns T_A into right multiplication by A^* on row vectors, and it fixes every e_i. So μ_A is also the measure attached to the integer matrix A^* in that convention.

Statements

Theorem 1.1. *There are a finitely generated discrete group G_D, an integer $n \geq 1$, and $A \in \mathrm{M}_n(\mathbb{Z}[G_D])$ invertible over $\mathbb{Q}[G_D]$ such that*

$$0 < \det_{\mathcal{N}(G_D)}(T_A) < 1.$$

The integral in Equation (1.1) is finite and strictly negative. Thus the unrestricted group-ring Determinant Conjecture has a negative resolution.

A unital ring is *directly finite* if $ab = 1$ implies $ba = 1$. The proof rests on one external result.

Theorem 1.2 ([8, Theorem 1.1])**.** *There exist a finite field K_2 of characteristic two, a finitely generated group H_2, and finite sums $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1$ and $b_2 a_2 \neq 1$.*

Unverified step. The whole result rests on Theorem 1.2, which is a counterexample to Kaplansky's direct finiteness conjecture for group algebras. The source does not prove or check it; it only cites the companion manuscript [8], which was not available for checking here. Every later step (Lemma 2.1 through Proposition 4.3 and the assembly in Section 5) is proved below, so Theorem 1.1 is conditional on this input.

Only this existence statement is used. The field is finite but not asserted to be $\mathbb{F}_2$. There is no torsion-free hypothesis (the companion construction uses torsion). Characteristic two is essential below. *Theorem 1.2 is cited, not proved or checked here; Theorem 1.1 is exactly as reliable as it is.* Everything after it is proved in full.

Proof at a glance

Key idea. Let $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$ have odd denominators, and suppose its reduction modulo 2 factors through r columns. Then for a suitable power q of two there is an *integer* matrix A of size $r + m$ with

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P)) \log q.$$

Up to two block shears, which do not change the determinant, A is the diagonal operator that multiplies r coordinates by q and the range of P by q^{-1}. A counterexample therefore needs a projection whose real trace exceeds its "rank modulo 2". A failure of direct finiteness in characteristic two supplies one.

Steps and dependencies.

(1) Lemma 2.1: turn a_2, b_2 into matrices X, Y over $\mathbb{F}_2[H_2]$ with $XY = I_d$ and a nonzero idempotent $C = I_d - YX$. Uses Theorem 1.2.

(2) Proposition 2.2: in $G_D = (\bigoplus_{H_2} C_3) \rtimes H_2$ build a projection P of size $d + 1$ with trace $d + \theta$, $\theta > 0$, and $\bar{P} = uv$ through d columns. Uses (1).

(3) Lemma 3.1: lift a factorization of $\bar{P}$ to one of P modulo $q = 2^k$. General G.

(4) Lemma 3.2: the matrix A is integral and equals $L_1 B R_1$ with B diagonal and L_1, R_1 shears. Uses (3).

(5) Lemmas 4.1 and 4.2: $\tau(\log(M^*M))$ is constant along $M(x) = L_x B R_x$.

(6) Proposition 4.3: the determinant formula. Uses (4), (5).

(7) Section 5: insert (2) into (6) with $r = d$, $m = d+1$, to get $\det = q^{-\theta}$.

2 From the scalar defect to a projection

All sums are finite, and matrix products keep the indicated order (the rings are noncommutative).

Lemma 2.1. *Let K_2 be a finite field of characteristic two and H_2 a group. For $d = [K_2 : \mathbb{F}_2]$, there is an injective unital ring homomorphism*

$$\rho : K_2[H_2] \longrightarrow \mathrm{M}_d(\mathbb{F}_2[H_2]).$$

For $a_2, b_2 \in K_2[H_2]$ with $a_2 b_2 = 1 \neq b_2 a_2$, their images X, Y satisfy $XY = I_d$ and $C := I_d - YX \neq 0$, with

$$C^2 = C, \qquad XC = 0, \qquad CY = 0. \tag{2.1}$$

This is used in Proposition 2.2, which needs coefficients in $\mathbb{F}_2$.

Proof. Step 1. Fix an $\mathbb{F}_2$-basis of K_2, and let $\mathrm{m}(\lambda) \in \mathrm{M}_d(\mathbb{F}_2)$ be the matrix of $x \mapsto \lambda x$. Then m is a unital ring homomorphism. It is injective because applying $\mathrm{m}(\lambda)$ to $1 \in K_2$ returns λ.

Step 2. Define $\rho(\sum_h \lambda_h h) = \sum_h \mathrm{m}(\lambda_h) h$, read as the matrix with (i,j) entry $\sum_h \mathrm{m}(\lambda_h)_{ij} h$. It is additive, and $\rho((\lambda h)(\mu k)) = \mathrm{m}(\lambda)\mathrm{m}(\mu)\, hk = \rho(\lambda h)\rho(\mu k)$ because scalar matrices commute with group elements. So ρ is a unital ring homomorphism. If $\rho(c) = 0$, then $\mathrm{m}(\lambda_h) = 0$ for each h, so $c = 0$ by Step 1.

Step 3. With $X = \rho(a_2)$, $Y = \rho(b_2)$: $XY = \rho(1) = I_d$, and $YX = \rho(b_2 a_2) \neq I_d$ by injectivity.

Step 4. $(YX)^2 = Y(XY)X = YX$, so $C^2 = I_d - 2YX + YX = C$. Also $XC = X - (XY)X = 0$ and $CY = Y - Y(XY) = 0$. □

The group G_D. Let H_2 act on $\bigoplus_{h \in H_2} C_3$ by left translation of coordinate labels, and let G_D be the semidirect product. Write z_h for a generator of the copy of C_3 labelled h. The z_h commute with each other and $s z_h s^{-1} = z_{sh}$ for $s \in H_2$. For coefficients in $\mathbb{Z}[1/3]$, a bar denotes the image under the ring homomorphism $\mathbb{Z}[1/3] \to \mathbb{F}_2$, applied coefficientwise. It exists because 3 is odd.

Proposition 2.2. *Let $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$, $d \geq 1$, satisfy $XY = I_d \neq YX$. In the group $G_D = (\bigoplus_{h \in H_2} C_3) \rtimes H_2$ there is an orthogonal projection $P \in \mathrm{M}_{d+1}(\mathbb{Z}[1/3][G_D])$ and a finite set $S \subseteq H_2 \backslash \{1\}$ such that*

$$\tau_{d+1}(P) = d + \theta, \qquad \theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad \bar{P} = uv,$$

where u has size $(d+1) \times d$ and v has size $d \times (d+1)$ over $\mathbb{F}_2[G_D]$. The integer $D = 3^{1+|S|}$ clears all coefficients of P.

Plan. Pick one nonzero coefficient of the defect C. Build a rational projection p that, sandwiched around the corresponding entry of C, keeps that one term and kills the others. Then $\mathrm{diag}(I_d, p)$ has trace $d + [1]p$, while modulo 2 the extra block $\bar{p}$ factors through C, hence through the same d columns as $I_d = XY$.

Proof. Put $C = I_d - YX$. By the computation in Step 4 of Lemma 2.1, which uses only $XY = I_d$, Equation (2.1) holds; and $C \neq 0$.

Step 1 (choose a coefficient). Choose an entry C_{ij} and $h_0 \in H_2$ whose coefficient in C_{ij} is nonzero, hence equal to 1 in $\mathbb{F}_2$. Then

$$h_0^{-1} C_{ij} = 1 + \sum_{s \in S} s \tag{2.2}$$

for a finite set $S \subseteq H_2 \setminus \{1\}$, possibly empty.

Step 2 (the projection p). In $\mathbb{Q}[G_D]$ put

$$p_h = \frac{1 + z_h + z_h^2}{3}, \qquad p = p_1 \prod_{s \in S} (1 - p_s).$$

Since $z_h^3 = 1$, $(1 + z_h + z_h^2)^2 = 3(1 + z_h + z_h^2)$, so $p_h^2 = p_h$. Inversion swaps z_h and z_h^2, so $p_h^* = p_h$. All p_h commute. Hence p is a product of commuting self-adjoint idempotents, so $p = p^* = p^2$. Each factor has denominator 3, so $3^{1+|S|}p$ is integral. (This commuting-coordinate device is that of [2, Lemmas 3.1 and 3.4].)

Step 3 ($psp = 0$ for $s \in S$). Conjugation by s is a ring automorphism of $\mathbb{Q}[G_D]$ sending p_h to p_{sh}, so $sps^{-1} = p_s \prod_{t \in S}(1 - p_{st})$. The product $p \cdot sps^{-1}$ consists of commuting factors, among them $(1 - p_s)$ from p and p_s from sps^{-1}, and $(1 - p_s)p_s = 0$. Therefore

$$psp = p\,(sps^{-1})\,s = 0. \tag{2.3}$$

The labels st need not avoid $\{1\} \cup S$; the argument does not use that.

Step 4 (trace of p). The labels in $\{1\} \cup S$ are pairwise distinct, since $1 \notin S$. In a direct sum, a product of powers of z_h over distinct labels is the identity only if every factor is. Hence the identity coefficient of a product of elements supported on distinct coordinates is the product of their identity coefficients. With $[1]p_1 = \frac{1}{3}$ and $[1](1 - p_s) = \frac{2}{3}$,

$$\tau_1(p) = [1]p = \frac{1}{3} \prod_{s \in S} \frac{2}{3} = \theta > 0. \tag{2.4}$$

(For $S = \emptyset$: $p = p_1$, $\theta = \frac{1}{3}$.)

Step 5 (reduce modulo 2). Reduction is a ring homomorphism, so $\bar{p}^2 = \bar{p}$ and $\bar{p}s\bar{p} = 0$ for $s \in S$ by Equation (2.3). Regard $\mathbb{F}_2[H_2] \subseteq \mathbb{F}_2[G_D]$. By Equation (2.2),

$$\bar{p}\, h_0^{-1} C_{ij}\, \bar{p} = \bar{p}\bar{p} + \sum_{s \in S} \bar{p}s\bar{p} = \bar{p}.$$

Step 6 (factorization). Let f be the row of length d with entry $\bar{p}h_0^{-1}$ at position i and zeros elsewhere. Let w be the column of length d with entry $\bar{p}$ at position j and zeros elsewhere. Then $fCw = \bar{p}h_0^{-1}C_{ij}\bar{p} = \bar{p}$. Define

$$P = \operatorname{diag}(I_d, p), \qquad u = \begin{pmatrix} X \\ fC \end{pmatrix}, \qquad v = \begin{pmatrix} Y & Cw \end{pmatrix},$$

of sizes $(d+1) \times (d+1)$, $(d+1) \times d$, $d \times (d+1)$. By Equation (2.1),

$$uv = \begin{pmatrix} XY & (XC)w \\ f(CY) & fC^2w \end{pmatrix} = \begin{pmatrix} I_d & 0 \\ 0 & fCw \end{pmatrix} = \operatorname{diag}(I_d, \bar{p}) = \bar{P}.$$

Finally $P = P^* = P^2$ by Step 2, $3^{1+|S|}P$ is integral, and $\tau_{d+1}(P) = d + [1]p = d + \theta$ by Equation (2.4). □

3 Lifting the factorization and preserving integrality

Now G is any group, $P \in \mathrm{M}_m(\mathbb{Q}[G])$, and D is an odd positive integer with DP integral. The coefficients of P lie in $\mathbb{Z}[1/D]$. For $q = 2^k$, D is invertible modulo q, so there is a ring homomorphism $\mathbb{Z}[1/D] \to \mathbb{Z}/q\mathbb{Z}$ extending reduction of integers. Applied coefficientwise it gives $P_q \in \mathrm{M}_m((\mathbb{Z}/q\mathbb{Z})[G])$. Composing with $\mathbb{Z}/q\mathbb{Z} \to \mathbb{F}_2$ gives $\bar{P}$.

Lemma 3.1. *Let G be a group, $m, r \geq 1$, D an odd positive integer, and $P \in \mathrm{M}_m(\mathbb{Q}[G])$ satisfy $P^2 = P$ and $DP \in \mathrm{M}_m(\mathbb{Z}[G])$. Suppose $\bar{P} = uv$ over $\mathbb{F}_2[G]$, where u and v have respective sizes $m \times r$ and $r \times m$. For $q = 2^k$, $k \geq 1$, the reduction P_q has a factorization $P_q = u_q v_q$ of the same sizes over $(\mathbb{Z}/q\mathbb{Z})[G]$.*

This is used in Lemma 3.2 to make one block integral. (The source writes e for P_q.)

Plan. Lift u, v arbitrarily. The lifted product, compressed by P_q, differs from P_q by an element with even coefficients, which is nilpotent modulo 2^k. A finite geometric series corrects it.

Proof. Put $\mathcal{R} = (\mathbb{Z}/q\mathbb{Z})[G]$. Since reduction is a ring homomorphism, $P_q^2 = P_q$.

Step 1. Choose any matrices u_0, v_0 over $\mathcal{R}$ reducing to u, v modulo 2, and set

$$Z = P_q u_0 v_0 P_q, \qquad W = P_q - Z.$$

Then $P_q W = W P_q = W$ and $P_q Z = Z P_q = Z$.

Step 2 ($W^k = 0$). Modulo 2, Z becomes $\bar{P}uv\bar{P} = \bar{P}^3 = \bar{P}$, so W becomes 0: every coefficient of every entry of W is even. Each coefficient of W^k is a sum of products of k such coefficients, hence divisible by $2^k = q$, hence zero in $\mathbb{Z}/q\mathbb{Z}$.

Step 3. Let $Q = P_q + W + \cdots + W^{k-1}$ (so $Q = P_q$ if $k = 1$). Since $Z = P_q - W$ and P_q acts as an identity on W, the sum telescopes: $QZ = P_q - W^k = P_q$.

Step 4. Set $u_q = QP_q u_0$ ($m \times r$) and $v_q = v_0 P_q$ ($r \times m$). Then $u_q v_q = QP_q u_0 v_0 P_q = QZ = P_q$. Only finite sums and products occurred. □

Choices and notation for the integral matrix. Assume in addition $P = P^*$. Fix $q = 2^k > 1$ with $q \equiv 1 \pmod{D}$. Such k exists because 2 is a unit modulo the odd integer D (take k a multiple of its order; if $D = 1$ any $k \geq 1$ works). Let U, V be any integer matrices reducing modulo q to u_q, v_q of Lemma 3.1. Define

$$J = -q^{-1}P - (I_m - P), \qquad B = \mathrm{diag}(qI_r, J),$$

and for real x

$$L_x = \begin{pmatrix} I_r & 0 \\ xU/q & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & xV/q \\ 0 & I_m \end{pmatrix}, \qquad M(x) = L_x B R_x.$$

Why this shape. Eliminating the off-diagonal blocks V, U of a block matrix with corner qI_r subtracts UV/q from the lower-right block. To be left with J, that block must be $UV/q + J$. The congruence $UV \equiv P \pmod{q}$ is what makes it integral, and the minus signs in J (invisible in J^*J) produce the difference $UV - P$.

Lemma 3.2. *Suppose $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$, an odd $D > 0$ clears its denominators, and $\bar{P} = uv$ factors through $r \geq 1$ columns. Let $q = 2^k > 1$ satisfy $q \equiv 1 \pmod{D}$. Choose arbitrary integral lifts U, V of the factors supplied by Lemma 3.1. Then*

$$A = \begin{pmatrix} qI_r & V \\ U & q^{-1}(UV - P - q(I_m - P)) \end{pmatrix} \in \mathrm{M}_{r+m}(\mathbb{Z}[G]) \tag{3.1}$$

has an inverse in $\mathrm{M}_{r+m}(\mathbb{Q}[G])$.

The proof also records Equation (3.2), used in Proposition 4.3.

Proof. Step 1 (integrality). Write $P + q(I_m - P) = qI_m + (1-q)P$. Since $D \mid q-1$ and DP is integral, $(1-q)P$ is integral, so this matrix is integral. Its reduction modulo q is $0 + (1-0)P_q = P_q$. The reduction of UV is $u_q v_q = P_q$ by Lemma 3.1. Hence $UV - P - q(I_m - P)$ is an integer matrix reducing to 0 modulo q, i.e. all its coefficients are divisible by q.

Step 2 (B is invertible). Using $P^2 = P$ and $P(I_m - P) = 0$,

$$(-q^{-1}P - (I_m - P))(-qP - (I_m - P)) = P + (I_m - P) = I_m,$$

and likewise in the other order. So $J^{-1} = -qP - (I_m - P)$ and $B^{-1} = \mathrm{diag}(q^{-1}I_r, J^{-1})$, both rational.

Step 3 (the path). Multiplying in order,

$$BR_x = \begin{pmatrix} qI_r & xV \\ 0 & J \end{pmatrix}, \qquad M(x) = L_x B R_x = \begin{pmatrix} qI_r & xV \\ xU & x^2 UV/q + J \end{pmatrix}. \tag{3.2}$$

Since $UV/q + J = q^{-1}(UV - P - q(I_m - P))$, we get $A = M(1)$.

Step 4 (inverse). L_x^{-1} and R_x^{-1} are obtained by negating the off-diagonal block. Thus $A^{-1} = R_1^{-1}B^{-1}L_1^{-1} \in \mathrm{M}_{r+m}(\mathbb{Q}[G])$. More generally $M(x)$ and $M(x)^{-1} = R_x^{-1}B^{-1}L_x^{-1}$ are polynomials in x with coefficients in $\mathrm{M}_{r+m}(\mathbb{R}[G])$. No relation such as $PU = U$, $VP = V$ or $V = U^*$ was used. □

Sanity check. For G trivial, $m = r = 1$, $P = 1$, $D = 1$, $q = 2$, $U = V = 1$: $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $|\det A| = 1 = q^{\,r - \tau_m(P)}$, matching Proposition 4.3.

4 The trace calculation

(Renamings to avoid clashes: the source's generic matrices C, D are Φ, Ψ here, its $W = I - N/c$ is Ω, and its shear blocks S, T are α, β.)

Let $\mathcal{A}_n$ be the operator-norm closure of $\{T_\Phi : \Phi \in \mathrm{M}_n(\mathbb{C}[G])\}$, a unital C^*-algebra. We identify Φ with T_Φ. On group-ring matrices:

- $\tau_n(\Phi\Psi) = \sum_{i,j,g} (\Phi_{ij})_g (\Psi_{ji})_{g^{-1}}$, which is symmetric under $(\Phi, i, g) \leftrightarrow (\Psi, j, g^{-1})$. So $\tau_n(\Phi\Psi) = \tau_n(\Psi\Phi)$.
- $|\tau_n(\Phi)| \le n\|\Phi\|$ and $\tau_n(\Phi^*) = \overline{\tau_n(\Phi)}$, from $\tau_n(T) = \sum_i \langle Te_i, e_i\rangle$.

Hence τ_n is a continuous linear functional on $\mathcal{A}_n$, and by density it is a trace there (*cyclicity*).

Lemma 4.1. *For a continuously differentiable path $M : [0,1] \to \mathcal{A}_n$ of invertible elements,*

$$\frac{d}{dx}\tau_n\big(\log(M(x)^*M(x))\big) = 2\,\mathrm{Re}\,\tau_n\big(M'(x)M(x)^{-1}\big).$$

This is the tracial logarithmic derivative of [4, Lemma 2], in the unnormalized trace; it feeds Lemma 4.2.

Proof. Put $N = M^*M$, a C^1 path with $N' = M'^*M + M^*M'$.

Step 1 (uniform bounds). Inversion is continuous, so $\|M(x)\|$ and $\|M(x)^{-1}\|$ are bounded on $[0,1]$. Since $\|\xi\| \le \|M^{-1}\|\,\|M\xi\|$, we get $N \ge \|M^{-1}\|^{-2} I$. So there are constants $0 < \delta < c$ with $\delta I \le N(x) \le cI$ for all x.

Step 2 (series). Let $\Omega = I - N/c$. Then $0 \le \Omega \le (1-\delta/c)I$, so $\|\Omega\| \le \rho := 1 - \delta/c < 1$. By functional calculus applied to $\log(ct) = \log c + \log(1-(1-t))$,

$$\log N = (\log c)I - \sum_{j\ge 1} \frac{\Omega^j}{j},$$

converging in norm, uniformly in x.

Step 3 (termwise derivative). $(\Omega^j)' = \sum_{i=0}^{j-1} \Omega^i \Omega' \Omega^{j-1-i}$, so $\|(\Omega^j/j)'\| \le \rho^{j-1} \sup_y \|\Omega'(y)\|$. The derivative series converges uniformly, so we may differentiate termwise, and the continuous linear τ_n passes through sum and derivative. By cyclicity each of the j terms of $(\Omega^j)'$ has trace $\tau_n(\Omega^{j-1}\Omega')$; commutation of N and N' is not assumed. With $\Omega' = -N'/c$,

$$\frac{d}{dx}\tau_n(\log N) = \frac{1}{c}\sum_{j\ge 1} \tau_n(\Omega^{j-1}N') = \frac{1}{c}\,\tau_n\big((I-\Omega)^{-1}N'\big) = \tau_n(N^{-1}N'),$$

using the Neumann series and $I - \Omega = N/c$.

Step 4. With $N^{-1} = M^{-1}(M^*)^{-1}$ and cyclicity, $\tau_n(N^{-1}M^*M') = \tau_n(M^{-1}M') = \tau_n(M'M^{-1})$. Also

$$\tau_n(N^{-1}M'^*M) = \tau_n(MN^{-1}M'^*) = \tau_n\big((M^*)^{-1}M'^*\big) = \tau_n\big((M'M^{-1})^*\big) = \overline{\tau_n(M'M^{-1})}.$$

The sum is $2\,\mathrm{Re}\,\tau_n(M'M^{-1})$. □

Lemma 4.2. *Let $r, m \ge 1$ and let $B \in \mathcal{A}_{r+m}$ be invertible. Let α, β have respective sizes $m \times r$ and $r \times m$ over the operator-norm closure of $\mathbb{C}[G]$. For $x \in [0,1]$, let*

$$L_x = \begin{pmatrix} I_r & 0 \\ x\alpha & I_m \end{pmatrix}, \qquad R_x = \begin{pmatrix} I_r & x\beta \\ 0 & I_m \end{pmatrix}.$$

Then $\tau_{r+m}(\log((L_xBR_x)^*(L_xBR_x)))$ is constant in x.

This is used in Proposition 4.3 with $\alpha = U/q$, $\beta = V/q$.

Proof. Let $M = L_xBR_x$, a polynomial path with $M^{-1} = R_x^{-1}B^{-1}L_x^{-1}$. By the product rule, keeping order, $M' = L_x'BR_x + L_xBR_x'$, so

$$M'M^{-1} = L_x'L_x^{-1} + (L_xB)\,(R_x'R_x^{-1})\,(L_xB)^{-1}.$$

Direct multiplication gives

$$L_x'L_x^{-1} = \begin{pmatrix} 0 & 0 \\ \alpha & 0 \end{pmatrix}\begin{pmatrix} I_r & 0 \\ -x\alpha & I_m \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \alpha & 0 \end{pmatrix}, \qquad R_x'R_x^{-1} = \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix}.$$

Both have zero diagonal entries, hence trace 0. By cyclicity the conjugate of $R_x'R_x^{-1}$ also has trace 0. So $\tau_{r+m}(M'M^{-1}) = 0$, and Lemma 4.1 gives derivative zero. □

Proposition 4.3. *Let G be a discrete group, $m, r \ge 1$, and let $P = P^* = P^2 \in \mathrm{M}_m(\mathbb{Q}[G])$. Suppose an odd $D > 0$ clears its denominators and $\bar{P} = uv$ with $u \in \mathrm{M}_{m,r}(\mathbb{F}_2[G])$ and $v \in \mathrm{M}_{r,m}(\mathbb{F}_2[G])$. For every $q = 2^k > 1$ congruent to one modulo D, the matrix A of Lemma 3.2 satisfies*

$$\tau_{r+m}(\log(T_A^*T_A)) = 2(r - \tau_m(P))\log q, \tag{4.1}$$

$$\log \det_{\mathcal{N}(G)}(T_A) = (r - \tau_m(P))\log q. \tag{4.2}$$

It is integral and rationally invertible, and its positive-square spectral measure has support in a compact subinterval of $(0,\infty)$.

Proof. Step 1. Integrality and the rational inverse are Lemma 3.2.

Step 2 (reduce to B). By Equation (3.2), $A = M(1)$ and $B = M(0)$, where $M(x) = L_x B R_x$ has shear blocks U/q, V/q and B is invertible (Lemma 3.2, Step 2). Lemma 4.2 gives $\tau_{r+m}(\log(T_A^* T_A)) = \tau_{r+m}(\log(B^* B))$.

Step 3 (compute at B). Since $P^* = P$, $J^* = J$; and $J^2 = q^{-2}P + (I_m - P)$ because $P(I_m - P) = 0$. For a projection, $\exp(tP) = I_m + (e^t - 1)P = e^t P + (I_m - P)$. So $J^2 = \exp(-2\log q \cdot P)$ with $-2\log q \cdot P$ self-adjoint, whence $\log(J^2) = -2\log q \cdot P$. Therefore

$$B^* B = \operatorname{diag}(q^2 I_r,\ q^{-2}P + I_m - P), \qquad \log(B^* B) = 2\log q \operatorname{diag}(I_r, -P),$$

with trace $2\log q\,(r - \tau_m(P))$. This is Equation (4.1); halving it gives Equation (4.2) by Equation (1.2).

Step 4 (spectrum). T_A and $T_{A^{-1}} = T_A^{-1}$ are bounded, so $\|T_A^{-1}\|^{-2} I \le T_A^* T_A \le \|T_A\|^2 I$, and μ_A is supported in $[\|T_A^{-1}\|^{-2}, \|T_A\|^2] \subset (0, \infty)$. By Equation (1.2) with $n = r + m$,

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = \frac{2(r - \tau_m(P))\log q}{r + m}. \tag{4.3}$$

□

5 Assembly and exact determinant

Proof of Theorem 1.1. Step 1. Take K_2, H_2, a_2, b_2 from Theorem 1.2 and set $d = [K_2 : \mathbb{F}_2] \ge 1$. Lemma 2.1 gives $X, Y \in \mathrm{M}_d(\mathbb{F}_2[H_2])$ with $XY = I_d \ne YX$.

Unverified step. This step uses the existence of K_2, H_2, a_2, b_2 from Theorem 1.2. That statement is a counterexample to Kaplansky's direct finiteness conjecture for group algebras, cited from the companion manuscript [8] and neither proved nor checked here. The conclusion of Theorem 1.1 is conditional on it.

Step 2. Proposition 2.2 gives $G_D = (\bigoplus_{H_2} C_3) \rtimes H_2$, an orthogonal projection P of size $d + 1$ with $\tau_{d+1}(P) = d + \theta$, a factorization $\bar{P} = uv$ through d columns, and

$$\theta = \frac{1}{3}\left(\frac{2}{3}\right)^{|S|} > 0, \qquad D = 3^{1+|S|} \text{ odd, clearing the coefficients of } P.$$

Step 3. Choose $k \ge 1$ with $q = 2^k \equiv 1 \pmod{D}$. Proposition 4.3 with $r = d$, $m = d + 1$ gives $A \in \mathrm{M}_{2d+1}(\mathbb{Z}[G_D])$, invertible over $\mathbb{Q}[G_D]$, with

$$\log \det_{\mathcal{N}(G_D)}(T_A) = (d - (d + \theta))\log q = -\theta \log q.$$

Since $q > 1$ and $\theta > 0$,

$$0 < \det_{\mathcal{N}(G_D)}(T_A) = q^{-\theta} < 1. \tag{5.1}$$

Step 4. With $n = 2d + 1$, Equation (4.3) reads

$$\int_{(0,\infty)} \log t \, d\mu_A(t) = -\frac{2\theta \log q}{2d + 1} < 0, \tag{5.2}$$

finite because μ_A has compact support in $(0, \infty)$ (Proposition 4.3).

Step 5. G_D is generated by a finite generating set of H_2 together with z_1, since $z_h = h z_1 h^{-1}$. □

6 Context and verification checklist

Context (not used in the proof). Determinant bounds belong to the theory of L^2-invariants [7]. They are known for large classes of groups, including amenable groups [9, Theorem 1.21, with its erratum] and sofic groups [3, Theorem 5]. The source remarks that the latter, applied to the positive integral matrix A^*A, shows G_D is not sofic; this is an external consequence, not an ingredient. The lower-bound question differs from determinant *approximation* along finite quotients, for which see [5, Theorem 2 and Section 2]. The projections p_h are related to the commuting wreath-product projections of [2, Section 3] and [6, Section 2]; no spectral theorem is imported from those works. The logarithmic derivative is classical [4, Lemma 2], see also [1, Lemma 1 and Proposition 2(d)]. The counterexample has no spectral mass at zero: it violates the lower bound, not the determinant-class property.

Verification checklist.

1. **External input.** Theorem 1.2 is taken from [8] and not checked here. Everything else is self-contained.
2. **Coefficient separation.** $psp = 0$ for $s \in S$, and $[1]p = \theta$: Proposition 2.2, Steps 3–4. Check that Step 3 needs only the single pair of factors $(1 - p_s)$, p_s, and Step 4 only that $1 \notin S$.
3. **Binary factorization.** $uv = \bar{P}$ with u of size $(d+1) \times d$: Proposition 2.2, Step 6. Check the order of the noncommutative products and the use of $XC = 0$, $CY = 0$, $C^2 = C$.
4. **Lifting.** $W^k = 0$ and $QZ = P_q$: Lemma 3.1, Steps 2–3. The mod-2 factorization and $2^k = q$ enter here.
5. **Integrality.** q divides $UV - P - q(I_m - P)$: Lemma 3.2, Step 1. This is where $q \equiv 1 \pmod{D}$ is needed.
6. **Shear invariance.** $\tau(M'M^{-1}) = 0$ along $M(x) = L_x B R_x$: Lemma 4.2, resting on Lemma 4.1 (Steps 3–4 use cyclicity without assuming N, N' commute).
7. **Endpoint value and normalization.** $\log(B^*B) = 2 \log q \operatorname{diag}(I_r, -P)$ and the factors $\frac{1}{2}$, $\frac{1}{r+m}$: Proposition 4.3, Steps 3–4, against Equation (1.2).

References

[1] Pierre de la Harpe and Georges Skandalis. Déterminant associé à une trace sur une algèbre de Banach. *Annales de l'Institut Fourier*, 34(1):241–260, 1984.

[2] Warren Dicks and Thomas Schick. The spectral measure of certain elements of the complex group ring of a wreath product. *Geometriae Dedicata*, 93:121–137, 2002.

[3] Gábor Elek and Endre Szabó. Hyperlinearity, essentially free actions and L^2-invariants. The sofic property. *Mathematische Annalen*, 332:421–441, 2005.

[4] Bent Fuglede and Richard V. Kadison. Determinant theory in finite factors. *Annals of Mathematics*, 55(3):520–530, 1952.

[5] Holger Kammeyer. A counterexample to the determinant approximation conjecture. arXiv:2609.15567v1, 2026. Version 1, September 14, 2026.

[6] Franz Lehner, Markus Neuhauser, and Wolfgang Woess. On the spectrum of lamplighter groups and percolation clusters. *Mathematische Annalen*, 342(1):69–89, 2008.

[7] Wolfgang Lück. *L^2-Invariants: Theory and Applications to Geometry and K-Theory*, volume 44 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*. Springer, 2002.

[8] OpenAI. A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two. OpenAI Math Release preprint OAI:A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Characteristic-Two-September-23-2026, 2026. Theorem 1.1 and Section 6.

[9] Thomas Schick. L^2-determinant class and approximation of L^2-Betti numbers. *Transactions of the American Mathematical Society*, 353(8):3247–3265, 2001. With the appended erratum in arXiv:math/9807032v2.